NUTRITION



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What is nutrition?

Nutrition is the science of how the body obtains and uses nutrients.

Nutrients are substances required for:

- Energy
- Growth
- Repair
- Maintenance of body functions

Nutrition strongly influences health and disease.

Why is nutrition important?

Good nutrition is essential for:

- Normal growth
- Brain function
- Energy production
- Immune function
- Hormone synthesis
- Enzyme activity
- Maintaining body weight
- Prevention of nutritional diseases

Nutrients are classified into:

- **Macronutrients:** needed in large amounts by the body e.g. Carbohydrates, proteins and fats.
- **Micronutrients:** needed in small amounts by the body e.g. Vitamins and minerals

Other essential component

- Water

Carbohydrates

1 gm provides 4 kcal of energy

Carbohydrates are major sources of energy.

Main function: provide energy, especially glucose for the brain and red blood cells

Sources

Carbohydrates mainly come from the food groups of grains, cereals, fruits, sweets etc.

Grains are a Carbohydrate

- Grains differ in size and shape, but they all have kernels with similar structure.
- Used to make a variety of foods like breakfast foods, flours, meals, breads, pasta products, and starches.



Nutritional Value of Grains

- Excellent source of complex carbohydrates
- Whole grain cereals contribute B vitamins, iron, magnesium, and zinc.
- Many processed products have nutrients added to them.

Proteins

1 gm protein = 4 kcal of energy

- Proteins are polymers of **amino acids**.
- Protein is one of the most important elements of our diets because it provides energy, builds and repairs tissue, and maintains body function.



Sources

- The best sources of protein for optimum health are all animal foods (milk, eggs, cheese, fish and red meat).
- Proteins also include legumes, dried beans, grains, peanuts, nuts, dried peas, beans, lentils and plants foods (wheat, corn, rice, etc.)

Fats

1 gm fat = 9 kcal of energy

Lipids are an important source of energy and structural components of cells.

- Store energy for future use
- Keep the body warm
- Protect organs
- Help absorption of vitamins A, D, E, and K

Fatty acids

Fatty acids can be classified as:

- **Saturated fatty acids** (No double bonds)
- **Unsaturated fatty acids** (One or more double bonds)

SOURCES

Good fat: olive oil, avocado, nuts, seeds, fish, Almonds, chia seeds

Bad fat: butter, fatty meat, full-fat dairy, fried foods, processed foods



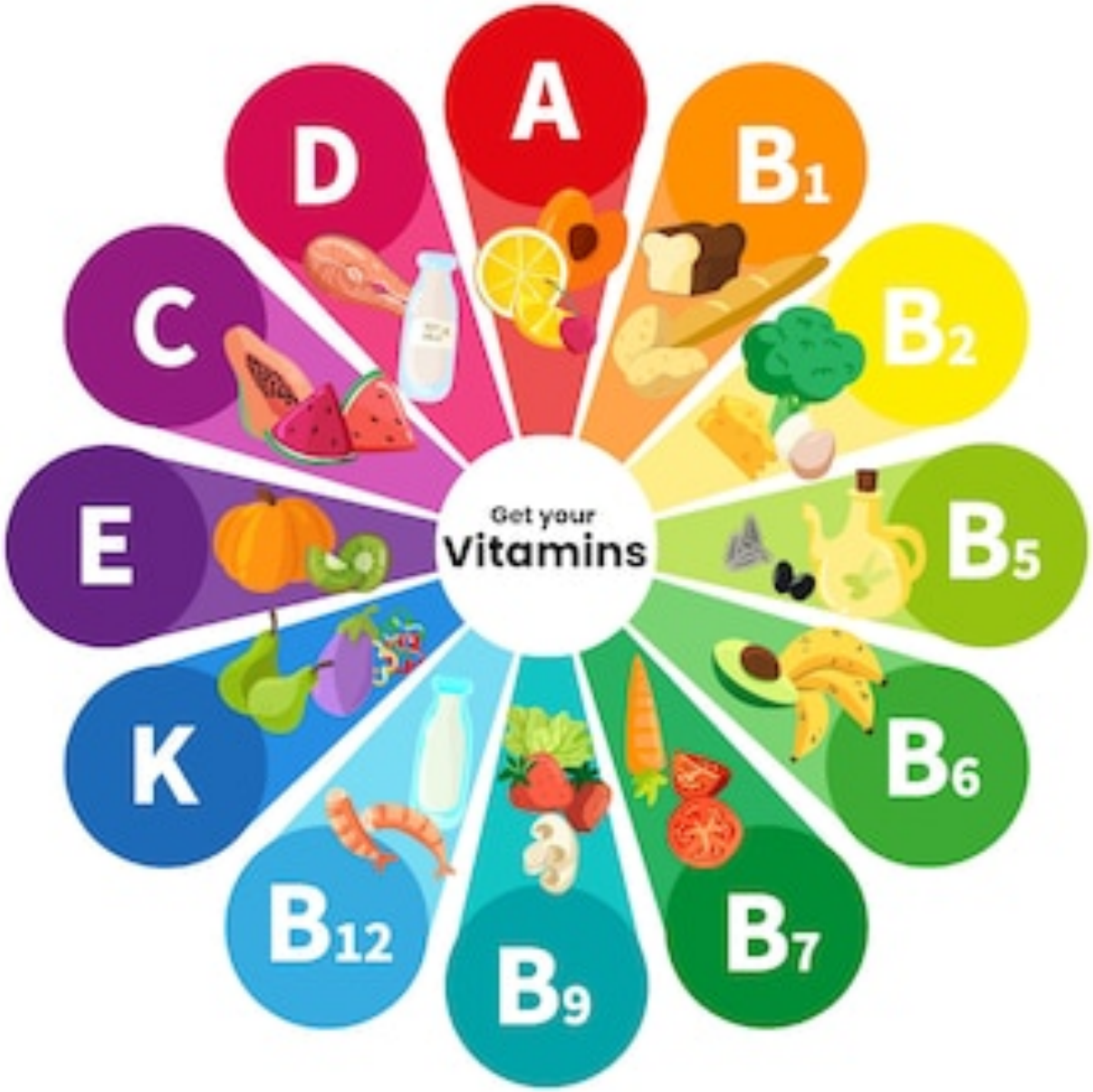
Vitamins

Vitamins are organic compounds required in small amounts.

They are essential for:

- Enzyme activity
- Metabolism
- Growth
- Immunity
- Cell function

- Two major groups of vitamin:
- **Fat-soluble:** A, D, E, K
- **Water-soluble:** B-complex and C



Vitamins (Fat soluble)

Vitamins	Function	Sources
Vitamin A	Prevention of nutritional blindness	carrots, liver, milk, green fruit, butter, fatty fish…….
Vitamin D	Prevention of rickets and helps formation of strong teeth	Fatty fish, egg, butter, cheese, full fat milk, and skin can produce vitamin D
Vitamin E	Acts as anti oxidant	Plant oils, whole grain cereals, egg, meat, green leafy vegetables.
Vitamin K	Helps blood clotting	Green leafy vegetables, cereals, legumes, milk, eggs, meat and fish

Vitamins (Water soluble)

Vitamins	Function	Sources
Vitamin C	Absorption of iron, bone formation, electron transport, etc	Tomatoes, guava, pine apple, ripe papaya and some green leafy vegetables like drumstick amaranth
Vitamin B1 (Thiamin)	Severe Deficiency leads to berberi, deficiency also weaken cellular function	Whole cereals, legumes, oilseeds and nuts, dried beans
Vitamin B2 (Riboflavin)	Deficiency leads to oral and facial lesions(damage to skin)	Liver, milk powder, eggs, meat and fish, whole cereals, legumes and green leafy vegetables
Vitamin B12 (Cobalamin)	Deficiency leads to deadly anaemia	All animal foods

Minerals

Minerals are Inorganic elements required for normal physiological function.

Most important is calcium growth and repair of bones and teeth

Dietary minerals come primarily from a wide variety of whole foods, including mineral-rich foods like nuts, seeds, lean meats, eggs, iodized table salt, water, liver, seafood, dairy, and leafy green vegetables.



Water

- Solvent for biochemical reactions
- Transport of nutrients
- Removal of waste products
- Temperature regulation
- Maintains blood volume
- Lubricates joints
- Saliva



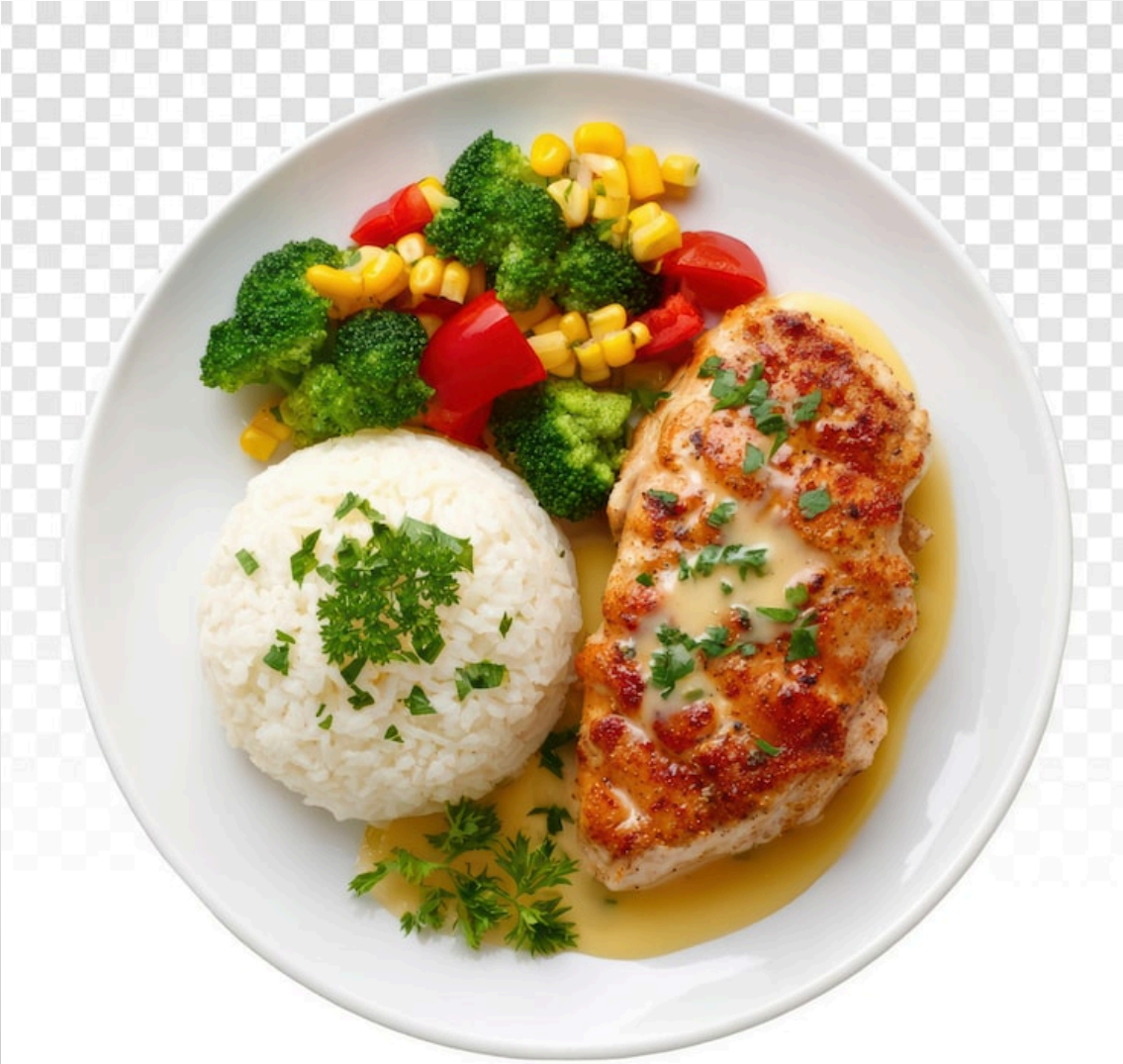
Nutritional Value of Dairy

- Excellent nutrition – fats, carbs, proteins, vitamins and minerals – riboflavin, vitamin B12, magnesium, potassium, and vitamins A and D
- Calcium rich – strengthens bones, helps body function properly
- Low in sugar unless processed into products such as ice cream



Balanced diet

Balanced diet: A diet that provides the right amounts and proportions of carbohydrates, proteins, fats, vitamins, minerals, fiber, and water to maintain good health and support normal body functions.



Balanced diet(cont.)

- For a balanced daily diet, a common guideline for total energy is:
- **Carbohydrates:** 45–65% of daily energy
- **Protein:** 10–35% of daily intake
- **Fat:** 20–35% of your diet
- These percentages refer to total daily calories, not grams.
- **Vitamins and Minerals:** Regulate body functions and boost immunity, found in fresh produce.
- **Fiber and Water:** Aid digestion and keep the body hydrated.

Example for 2,000 kcal/day:

- **Carbohydrate: 50% = 1,000 kcal**
- **Protein: 20% = 400 kcal**
- **Fat: 30% = 600 kcal**

- **Remember:**
- carbohydrate & protein = 4 kcal/g;

  fat = 9 kcal/g.



Standard food-composition examples

Food	Protein (%)	Carbohydrate (%)	Fat (%)
Cooked white rice	2.5%	28%	0.3%
Bread	9%	50%	3%
Cooked potato	2%	17%	0.1%
Whole milk	3.2%	4.8%	3.3%
Egg	13%	1%	12%
Chicken	27%	0%	5%
Beef	26%	0%	15%
Lentils, cooked	9%	20%	0.4%
Banana	1.1%	23%	0.3%

For example, 250 g cooked rice:

- **Protein:**
  2.5% × 250 g = 6.25 g
  6.25 × 4 = **25 kcal**
- **Carbohydrate:**
  28% × 250 g = 70 g
  70 × 4 = **280 kcal**
- **Fat:**
  0.3% × 250 g = 0.75 g
  0.75 × 9 = **6.75 kcal**
- **Total = 25 + 280 + 6.75 = 311.75 kcal ≈ 312 kcal**

Nutrient	Approximate daily need per 1 kg body weight
Protein	0.8 g/kg/day
Carbohydrate	3–5 g/kg/day
Fat	0.8–1.2 g/kg/day
Fiber	~0.4 g/kg/day
Water	30–35 mL/kg/day

Example: 60-kg adult

- **Protein: 0.8 × 60 = 48 g/day**
- **Carbohydrate: 3–5 × 60 = 180–300 g/day**
- **Fat: 0.8–1.2 × 60 = 48–72 g/day**
- **Fiber: ~0.4 × 60 = 24 g/day**
- **Water: 30–35 × 60 = 1,800–2,100 ml/day**


Thank You!